uPVC ALUMINIUM WINDOW & DOOR MACHINERY

CATALOGUE COLLECTION

PRECISION ENGINEERING

PRL
Precision.Reliability.Loyalty

PARIDA RED LION INDIA PVT LTD

Real Make in India inspired by Germany Technology

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DIRECTOR MESSAGE

From Small Village Of Odisha To Delhi From Nothing To Prl

Prl's story began not with capital or comfort, but with courage, belief, & an unshakable dream

EVERY SUCCESS STORY HAS A BEGINNING EVERY COMPANY FOUNDER HAS A STORY.

Mine began in a small corner of Odisha After finish My engineering Study , There was a time when I started this journey with a heart full of dreams and pockets completely empty with no money, no support, and no clear path — only determination.

The early days were filled with challenges that tested my body and spirit. There were nights I worked without sleep, and days survived without food. There were moments when negativity surrounded me, people doubted me, and circumstances pushed me to give up . I Survived Only determination to change my life and create something meaningful.

- But the road ahead was harsher than I ever imagined.
- A Life of Extreme Hardship
- When I reached Delhi, life showed me its toughest side:

- There were days when I walked 30 kilometers because I didn't have even ₹10 for a metro ticket or a rickshaw.
- For four days, I survived without food, fighting hunger, weakness, and loneliness.
- Due to severe vitamin deficiency, my body collapsed — I became paralysed.
- I had no one to take care of me. I admitted myself to the hospital… completely alone.
- I slept on the cold winter ground without any bedsheet and pillow because I had no home.
- I faced betrayal from people I trusted, again and again.
- I couldn't go back home because I came from a poor family, and returning meant giving up on my dreams.

One day, after saving little by little, I bought a kambal (blanket) with my own money to protect myself from the winter cold.
That day…was the happiest day of my life. Not because it was a blanket…
but because for the first time, I felt I had won against life. I felt safe. I felt strong. I felt that my struggle was slowly turning into strength.

My entire life was a test a long, painful struggle filled with moments that could break any human. But each time life pushed me down, I stood up stronger. With every setback, I reminded myself why I started: to build something meaningful, something world-class, something that proves that struggle is not a weakness, it is a foundation

Turning Pain into Purpose

When there was no one beside me, courage became my only partner.
When the world doubted me, self-belief became my strength.When I had no resources, determination became my fuel.

Slowly, I started working… learning… building… failing… and rising again.

PRL wasn't created in comfort was created in pain, in hunger, in cold nights, in silent tears, in endless effort , With time, my small steps transformed into a vision.
That vision became a mission. And that mission became PRL – Parida Red Lion India Pvt. Ltd.
Those difficult years shaped the PRL you see today. PRL one of India's leading manufacturers in the uPVC & Aluminium Machine Industry. stands as proof that perseverance beats obstacles and that Indian engineering can meet and exceed global standards. Every machine we deliver carries not just technology, but also a part of my journey

- We operate from India's largest 28,000 sq. ft World-Class Manufacturing Infrastructure in uPVC Machine Manufacturing industry .
- We have our own World-Class Machine Tool Room, equipped for precision and innovation.
- We are a strong family of 50+ dedicated employees.
- We proudly serve 500+ clients across PAN India.
- Our machines are known for their unique design, advanced technology, and unmatched reliability making PRL one of the most trusted and loved brands in our industry.

What began from pain is today shaping industries & empowering fabricators across the country. Today, PRL stands the belief that no dream is too big, & no struggle is too small to matter. Today, when you see our modern machines, our factory, our team, and our global presence — remember this:

Behind every machine...
behind every innovation...
behind every success...
lies a journey of survival.

PRL – Built on Values Born From Life

My struggles shaped the core values of PRL:

Innovation – because when you have nothing, creativity becomes your strongest weapon.
Customer Delight – because people who trust you deserve honesty, respect, and excellence.
Sustainability – because lasting success must protect the future.
Excellence – because quality is the only way to rise.
Global Reach – because dreams should have no boundaries.

A Message from the Founder

If you ever feel like giving up, remember this:

I walked 30 km with an empty stomach…
I slept on the ground in winter Without Blanket …
I fought paralysis alone…
I survived hunger, betrayal, and poverty.
If I can rise from nothing…
you can rise too. "Its your time".

PRL is not just a company.
It is proof that no dream is too big and no background is too small.
To our customers, partners & my dedicated team thank you for trusting our vision and becoming part of this incredible journey.

From nothing to something this is our story and it has only just begun.

ABOUT US

PARIDA RED LION INDIA PVT LTD (PRL) is a leading manufacturer of uPVC Aluminium window making machines, dedicated to delivering innovative, high-quality solutions for the uPVC & Aluminium window & door fabrication industry. Established in 2017 & Entered uPVC & Aluminium Machine Manufacturing Facility in 2022 , our company is Manufacturing Facility Located in Greater Noida & we pride ourselves on our commitment to excellence, precision engineering & customer satisfaction.

By using the newest technologies that are discovered for the area of uPVC & Aluminium window & door production, PRL uPVC & ALUMINUM Window Making Machine constantly improves & updates the window & door manufacturing equipment & window machines designs according to market requirements. The uPVC & Aluminium window Making machines assortment of the company includes various models of uPVC window machines & aluminium window machines especially built for a smooth & reliable window manufacturing experience.

With over 8yr years of experience, we specialize in designing and producing a wide range of uPVC & Aluminium window manufacturing machines, including uPVC Aluminium Window Making cutting saws Machine , uPVC Corner welding machines, CNC corner cleaning , and auxiliary equipment like water slot milling and glazing bead cutting machines. Our state-of-the-art production facilities, spanning 10,000 sqr ft , are equipped with advanced technology to ensure superior performance, durability, and efficiency in every machine we create.

Our mission is to empower uPVC & Aluminium window and door manufacturers with reliable, cost-effective, and user-friendly machinery that enhances productivity and meets diverse fabrication needs. Our uPVC & Aluminium Window Making Machine machines are designed for ease of use, low maintenance, and energy efficiency, helping our clients reduce operational costs while maintaining high-quality output

At PRL , we prioritize innovation and sustainability. Our dedicated research and development team continuously works to improve our products, incorporating the latest advancements in automation and precision engineering. We are committed to environmentally responsible practices, ensuring our machines minimize waste and energy consumption.

We are proud to serve a Pan India & Globe clientele, with a strong presence. Our comprehensive after-sales support, including training, spare parts availability, and technical assistance, ensures our customers achieve maximum uptime and operational success., we adhere to the highest industry standards, delivering machines that are trusted worldwide for their reliability and performance.

Choose PARIDA RED LION INDIA PVT LTD (PRL) as your trusted partner for uPVC window making machinery, and let us help you build the future of window and door fabrication with precision, efficiency, and innovation. The special side of the company is that uPVC & Aluminium window machinery are sold at affordable prices, which is one of its advantages of being the manufacturer and seller at the same time. This allows to purchase high quality uPVC & Aluminium machines door manufacturing equipment and uPVC & Aluminium window machinery not only for large companies but also for small companies as well.All production has guaranteed high quality proven by long-term experience of successful operation in a world wide market of window machinery and door machinery.

Who We Are:

Established in 2017 We started with nothing no funding, no facility, not even a proper workspace. Just a vision. A big one. What we lacked in money, we made up for in determination, grit, and hard work. From a small 4 person sitting office building our first prototypes with basic tools, every step of our journey was powered by belief that one day, we'd build something that matters.

Today, that belief has taken shape. We now operate a full-fledged manufacturing facility of 10000sqrft , equipped with advanced machinery and run by a dedicated team of over 50 full-time professionals. From humble beginnings to a growing name in the machine manufacturing industry, our focus has never wavered: quality, innovation, and customer trust.

has grown into a reputable name in the uPVC machinery sector. Our journey began with a deep understanding of the evolving demands of the window fabrication industry and a dedication to addressing these needs through robust and intelligent machine solutions. We pride ourselves on our team of experienced engineers, designers, and technical experts who continuously push the boundaries of what's possible in uPVC machinery

We design and build high-performance machinery that empowers industries from aluminium and uPVC fabrication to advanced automation solutions. Every machine that leaves our floor carries the spirit of where we came from and where we're headed.
We are not just a company. We are proof that big dreams, backed by hard work, can build something real. Something lasting.

What We Offer:

We specialize in the complete lifecycle of uPVC machine manufacturing, from conceptualization and design to fabrication, assembly, testing, and deployment.
Our core offerings typically encompass:, We specialize in designing, developing, and manufacturing a comprehensive range of uPVC window and door making machines, including:

- **uPVC &Aluminium Window Making Cutting Machines:** Precision single and double-head cutting saws for accurate profile preparation.
- **Welding Machines:** State-of-the-art single, double, triple, and four-head welding machines for strong & seamless frame fabrication.
- **Corner Cleaning Machines:** Automated and manual corner cleaning solutions for a perfect finish.
- **Milling Machines:** Water slot, end milling, and other specialized milling machines for various profile operations.
- **Glazing Bead Saws:** Dedicated machines for precise cutting of glazing beads.
- **Reinforcement Screw Fixing Machines:** Efficient solutions for securing steel reinforcements.
- **Ancillary Equipment:** A full suite of supporting machinery and tools for a complete production line.

Why Choose PRL?

Our Commitment to Quality & Innovation:

At PRL , quality is not just a standard; it's our foundation. Every machine is meticulously engineered using high grade materials & components, undergoing rigorous testing to ensure durability, reliability & optimal performance.

We continuously invest in research and development, incorporating the latest technological advancements such as PLC control systems, servo motors, and intuitive HMI interfaces to enhance automation, accuracy, and ease of use for our clients.

We adhere to stringent quality control protocols at every stage of the manufacturing process, from raw material sourcing to final product inspection. Our machines are built to withstand rigorous industrial environments, ensuring exceptional durability, minimal downtime & consistent performance.

Innovation is embedded in our DNA. We continuously invest in research and development, leveraging the latest advancements to enhance our product capabilities. Our aim is to deliver machines that are not only efficient & reliable but also intelligent, user-friendly & future-proof.

- Expertise: Decades of collective experience in uPVC machinery design and manufacturing.
- Reliability: Machines built for continuous, heavy-duty operation with minimal downtime.
- Precision: Advanced technology ensuring high accuracy and consistent product quality.
- Customization: Solutions tailored to meet specific production requirements and capacities.
- Comprehensive Support: Excellent pre-sales consultation, installation, training & prompt after-sales service.
- robust logistical support.

Partner with Us:

Whether you are setting up a new uPVC window manufacturing unit or looking to upgrade your existing machinery, PRL is your ideal partner. We are dedicated to helping you achieve higher productivity, superior product quality, and greater profitability.

QUOTATION

PRODUCT CLIENT ORDER FORM

Quotation No – PRL /QU/24/25..... **Date –**

Dealing Person -

1. Client Information

Field	Details
Client Name	
Company Name (if any)	
Contact Number	
Email Address	
Address	
GST	

PRODUCT DETAILS

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2. Order Details

Sl.No	Product Name	Product Code	Final Price
1.			
2.			
3.			
4.			
5.			
6.			
7.			
8.			
9.			
10.			

Order Total:

GST % :

Total Value -

Extra Accessories & Die

1- Welding block

2- Beading block

3- End milling Cutting

4- Cleaning Electrical Hand Tool

5- Drainage Electrical Hand Tool

6- Pneumatic Hand Tool Set

7- Mechanical Cleaning Machine

8- Air Compressor

9- Air Dryer

10- Transport

3. Payment Terms

1. Payment Method: Bank Transfer ☐ , Cash ☐

2. Payment Terms:

Full payment upon order confirmation.

PRODUCT PAGE

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1. DOUBLE HEAD uPVC SEAMLESS WELDING MACHINE (DLD600)

Product Overview

A PRL Make uPVC Double Head Seamless Welding Machine is specifically designed to produce uPVC (Unplasticized Polyvinyl Chloride) windows & doors. This machine is used to weld two uPVC profiles together. It has two welding heads that simultaneously apply heat & pressure to the profiles through a PLC program, melting them together & creating a seamless bond. It is engineered to provide precision, durability & seamless finishes, ensuring high-quality results that meet modern construction standards.

Key Benefits

- Seamless welds with removal of excess material for a clean, invisible seam
- Dual-head operation doubles throughput
- Flexible welding angles (45 & 90 Degree)
- PLC control with touchscreen interface for repeatable, high-quality results
- Robust, low-maintenance construction with energy-efficient operation
- Comprehensive after-sales support & training

Parameter	Specification
Machine Type	Double Head Seamless Welding Machine
Welding Material	UPVC profiles
Welding Heads	2 heads (dual-head operation)
Welding Width	0–120 mm
Welding Height	20–180 mm
Welding Angle Range	45°–90° (models vary; 45° highlighted)
Heating Time	Adjustable (typical 0–300 s)
Cooling Time	Adjustable (typical 0–300 s)
Power Supply	220V/380V, 50 Hz (customizable)
Heating Power	1.5 kW per head
Air Pressure	0.6–0.8 MPa (external air compressor required)
Clamping	Automatic pneumatic
Positioning Accuracy	±0.2 mm
Control System	PLC with touchscreen interface
Operation Panel	Touchscreen & button control
Dimensions	~4500 × 1000 × 1700 mm
Weight	~0.8 ton
Production Capacity	~100–120 frames per 8-hour shift
Safety Features	Emergency stop, overload protection
Warranty	1 years

2. SINGLE HEAD SEAMLESS WELDING MACHINE - SLD 300

The PRL MAKE UPVC Single Head Seamless Welding Machine is a cutting-edge solution for precision welding of UPVC windows and doors. It is ideal for smaller production lines or custom applications, delivering seamless joints, superior aesthetics, and unmatched durability to meet modern quality standards.
The UPVC Window Making Single Head Seamless Welding Machine automates the welding process for UPVC window profiles, combining precision engineering with user-friendly features for consistent, high-quality welds, durability, and seamless performance.

Key Benefits

- Single Head Welding System: Accurate, seamless corner welds with minimal finishing.
- Seamless Welding Technology: Gap-free joints foraesthetics, strength, insulation; no extra post-processing.
- Automatic Control System: PLC-based, adjustable temperature, pressure, and time for different UPVC profiles.
- High Welding Precision: Precise alignment and clamping for uniform welds.
- Energy Efficient: Optimized heating minimizes costs without sacrificing speed/quality.

- Durability and Robust Construction: Built for continuous production, low maintenance.
- User-Friendly Interface: Intuitive touch-screen with real-time data and diagnostics.
- Customizable Profile Clamping: Adjustable clamps reduce deformation; flexible for various profiles.
- Safety Features: Sensors and emergency stop; complies with international safety standards.

Parameter	Details
Power Supply	220V/380V, 50Hz
Welding Range	Profiles from 20mm to 160mm width
Welding Angle	30° to 180°
Heating Time	0-300 seconds (adjustable)
Welding Time	0-180 seconds (adjustable)
Control System	PLC-based with digital display
Machine Dimensions	Approx. 1200mm x 900mm x 1700mm
Weight	Approx. 400 kg
Air Pressure Requirement	0.5-0.8 MPa
Power Consumption	1.5 kW

3. SINGLE HEAD UPVC V-WELDING MACHINE (SVD-300)

A single-head V-welding machine is an essential & foundational piece of equipment in the fabrication of UPVC (Unplasticized Polyvinyl Chloride) windows. Its primary and most critical use is to weld the corners of UPVC window profiles, The UPVC window single head V welding machine is primarily used for welding V-shaped ,corner joints & UPVC window & door profiles, creating seamless, strong connections at the corners of window frames

Core Function: Welding UPVC Window Corners

1.Creating Window Frames:

- UPVC window frames are typically constructed from extruded profiles that are cut to specific lengths and angles.
- The single-head V-welding machine joins these cut profiles together at the corners, usually at 45-degree angles, to form the rectangular or square shape of the window frame and sash.

2.Achieving Watertight and Airtight Seals:

- The welding process melts the surfaces of the UPVC profiles, allowing them to fuse together molecularly.
- This creates a continuous, monolithic corner joint that is essential for preventing water ingress and air leakage around the window.

3.Ensuring Structural Integrity:

- The fused weld provides the necessary strength and rigidity to the window frame, allowing it to withstand the forces it will encounter during installation and throughout its lifespan.

4.Producing Clean, Aesthetic Corners:

- The "V" shape often refers to the way the machine holds & welds the profile, often creating a clean, sharp corner. Specialized welding bars and molds ensure that the exterior appearance of the window frame is neat and free from excess material (weld burr), which is then typically cleaned off by a subsequent corner-cleaning machine.

Parameter	Specification/Value
Model	VLD -300
Power Supply	380V / 220V, 50Hz
Heater Power	1.2 kW
Welding Length	400–3300 mm
Profile Height	20–160mm
Profile Width	≤120 mm
Welding Angle Range	45°–180°
Heating Temperature	0–399°C (adjustable)
Operation Air Pressure	0.4–0.6 MPa (4–6 bar)
Air Consumption	100–200 L/min
Control System	PLC,
Welding Seam Thickness	0.2 mm (standard),
Machine Weight	Approx 500 kg
Machine Dimensions	Approx. 1100×800×1700 mm
Min. Weldable Frame Size	400×400 mm
Max. Profile Height	Up to 160 mm
Max. Profile Width	Up to 120mm
Heating Plate Material	Aluminium
Clamping System	Pneumatic
Special Features	Adjustable supporting arms, quick-detachable molds, size tolerance compensation

4. SINGLE HEAD 500MM SAW ALUMINIUM CUTTING MACHINE (CAS-500)

The Single Aluminium & UPVC Heavy Duty 500 Cutting Machine is a robust and precise power saw designed for accurately cutting aluminium and UPVC profiles used in the fabrication of windows, doors, and other architectural elements. The "500" in its name typically refers to the diameter of the saw blade in millimeters, indicating a substantial cutting capacity. "Heavy Duty" signifies its construction for demanding industrial use, ensuring durability and stability.Its primary function is to deliver clean, straight, and precise cuts across various profiles, minimizing material waste and ensuring that components fit perfectly during assembly

Key Uses and Applications

1.Accurate Profile Cutting:

- The machine is used to cut aluminium and UPVC extrusions to exact lengths required for window frames, door frames, transoms, mullions, and other structural components.

2. Straight and Square Cuts:

- The machine is used to cut aluminium and UPVC extrusions to exact lengths required for window frames, door frames, transoms, mullions, and other structural components.

3. Compound Mitre Cuts (Common for Blade Diameter 500mm):

- Many machines of this size and type are equipped with a tilting and/or rotating saw head. This allows for precise compound mitre cuts essential for creating frames where corners are not just at 45 or 90 degrees but also on a slope, common in complex window designs or to accommodate specific installation angles.

4. Cutting Various Profile Shapes:

- Its heavy-duty design and larger blade capacity allow it to cut through thicker, wider, or more complex aluminium & UPVC profiles with ease.

5. High-Volume Production:

- The robust construction and efficient cutting mechanism make it suitable for manufacturing environments that require high throughput and consistent results.

Parameter	Specification
Machine Type	Single Head Heavy Duty Cutting Machine
Cutting Capacity	Up to 175 mm
Cutting Angles	90°, 45°
Motor Power	2.2 kW
Saw Blade Diameter	500 mm
Saw Blade Speed	2,800 RPM
Power Supply	380V/415V, 3 Phase, 50/60 Hz
Cutting Accuracy	± 0.1 mm
Angular Accuracy	± 0.1°
Machine Weight	600 kg
Machine Dimensions	1500 × 1000 × 1200 mm
Clamping System	Pneumatic

5. SINGLE HEAD UPVC & ALUMINIUM TOP TO BOTTOM CUTTING MACHINE (TCM-400)

The uPVC Top & Bottom 400mm Saw Cutting Machine is a high-precision Single blade cutting system designed to cut UPVC (rigid unplasticized polyvinyl chloride) profiles from Top To Bottom Movement surfaces in a single pass. The machine features Single 400mm diameter saw blades positioned vertically , enabling chip-free, burr-free cuts with professional-grade edge quality suitable for immediate assembly without post-processing.

Applications

1. **Precision Cutting of UPVC Profiles:**
   - Used for accurate cutting of top To bottom sections of UPVC window and door profiles, typically up to 120mm in width .
2. **Window and Door Frame Fabrication:**
   - Ensures correct sizing of frame rails for smooth assembly and fitting of hardware.
3. **Mass Production Efficiency:**
   - Manual cutting for Medium -volume production lines, reducing errors and increasing throughput.

4. **Clean and Burr-Free Edges:**

- Produces smooth cuts essential for weather sealing and joint strength.

5. **Versatile Profile Compatibility:**

- Suitable for various UPVC profile shapes and thicknesses used in residential and commercial fenestration.

Technical Parameter

Technical Parameter	Specification/Value
Cutting Length Capacity	Up to 400 mm
Cutting Thickness	2 mm – 4 mm (profile wall thickness)
Saw Blade Diameter	400 mm
Motor Power	1.5 kW
Cutting Speed	2800 RPM
Power Supply	220V
Cutting Angle	90° fixed / 45
Machine Dimensions (LxWxH)	Approx. 1200×800×1400 mm
Machine Weight	200kg
Operation Type	Manual
Safety Features	Blade guard, emergency stop

6. DOUBLE HEAD SEMI-AUTOMATIC CUTTING MACHINE 450MM SAW (ROXX-600)

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The is PRL Make uPVC Double Head Cutting Machine (Semi-Automatic) equipped with a 450MM saw blade is the ultimate solution for precision and efficiency in uPVC window and door manufacturing. Designed for professional-grade performance, this cutting machine offers unmatched accuracy, ease of operation, and durability, making it a vital asset for industries handling uPVC profiles and similar materials.

Primary Applications

1. Window and Door Profile Cutting The machine cuts UPVC and aluminium profiles for door and window manufacturing, including corner connectors and curtain wall component
2. Angle Cutting Equipped with automatic tilting heads for 45° & 90° angle cuts, with manual adjustment available for intermediate angles. At 90 degrees, the effective cutting height reaches 145mm
3. Double-Sided Simultaneous Cutting Features two 450mm saw blades that can operate independently, allowing for simultaneous cutting of multiple profiles or separate operations depending on production needs.

4 High-Precision Fabrication Delivers dimensional accuracy of ±0.1mm and angle accuracy of ±0.5 degrees, ensuring components fit perfectly during assembly.

5. Extended Cutting Lengths Supports cutting lengths from 450mm to 4500mm (up to 6000mm with software support on some models), accommodating various window & door size.

OPTIONAL ALUMINIUM –

- Coolant spray unit
- Hydro Pneumatic Cylinder
- Saw blade According to Aluminium
- Heavy mechanical Body Structure
- Motor Configuration

Technical Specifications

Parameter	Specification
Machine Type	Semi-Automatic Double Head Cutting Machine
Blade Diameter	450MM
Cutting Range	Up to 145mm in width, suitable for uPVCprofiles
Cutting Angles	45° to 90° adjustable
Motor Power	2 x 3 kW
Control System	Semi-Automatic
Material Compatibility	uPVC, Aluminum, PVC
Cutting Accuracy	± 0.1 mm
Power Supply	380V, 50Hz
Machine Dimensions	3200mm x 1200mm x 1800mm
Weight	Approx. 1050 kg
Safety Features	Emergency stop, blade guard overload protection

7. DOUBLE HEAD SEMI AUTOMATIC 500MM SAW ALUMINIUM CUTTING MACHINE (ROXX-900)

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The Semiautomatic 500mm Aluminium Double Head Cutting Machine is a specialized industrial saw designed for accurately and efficiently cutting aluminium profiles used in the manufacturing of windows, doors, curtain walls, and other architectural systems. The "500mm" refers to the diameter of the saw blades, and "Double Head" indicates it has two independently or simultaneously operable cutting heads. "Semiautomatic" signifies that while the cutting action is automated to some degree, manual intervention is required for profile positioning and initiating the cut.

Primary Uses and Applications

1. **Precision Mitre and Angle Cutting:**

. 45° Cuts: The primary function is to cut aluminium profiles at precise 45-degree angles for the corners of window & door frames. This allows for perfect joinery when two pieces are brought together.

. 90° Cuts (Square Cuts): It can also perform accurate 90-degree cuts for sections like transoms, mullions, or straight frame members.

2. **Simultaneous Cutting:**

• The double-head design allows both heads to move inwards simultaneously to cut both ends of a single profile in one operation. This is highly efficient for producing identical lengths for frame

3. **Cutting Various Aluminium Profiles:**

• Its robust construction and 500mm blades are designed to handle a wide range of aluminium extrusions, from thin-walled window profiles to thicker structural sections used in commercial facades.

4. **High-Volume Production:**

• The combination of simultaneous cutting, precise length control, and efficient operation makes it ideal for medium to high-volume manufacturing environments for fenestration products.

5. **Minimizing Material Waste:**

• The precision of the cuts reduces errors and the need for rework, optimizing material usage and improving cost-effectiveness.

6. **Ensuring High-Quality Finish:**

• Equipped with high-quality, sharp carbide-tipped blades and precise head movement, it delivers clean, smooth, and burr-free cuts, improving the aesthetic finish of the final product and facilitating subsequent assembly.

Component	Specification
Main Motor Power	3.0 kW
Motor Type	Three-Phase Induction Motor
Motor Speed	2800 RPM
Pneumatic System Pressure	0.6 - 0.8 MPa (6-8 Bar)
Air Consumption	40 L/min
Control System Voltage	24V DC
Total Power Consumption	3.5 kVA

Performance Parameter	Specification
Cutting Angle Range	45°- 90° (Both Directions)
Standard Angle Stops	90°, 45°,
Cutting Accuracy	±0.1 mm
Angular Accuracy	±0.1°
Repeatability	±0.05 mm
Cutting Cycle Time	8-15 seconds per cut
Production Capacity	400-600 ts per 8-hour shift

8. DOUBLE HEAD AUTOMATIC 450MM UPVC & ALUMINIUM CUTTING MACHINE (ROXX-600A)

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An Automatic CNC 450mm Double Head Cutting Machine is a highly advanced piece of equipment designed for high-precision, high-volume cutting of both UPVC and aluminium profiles. The "CNC" (Computer Numerical Control) designation signifies automated operation controlled by computer programs, while "450mm" refers to the blade diameter, "Double Head" indicates two cutting heads, and the ability to cut both UPVC and aluminium makes it versatile.
Its primary use is in high-efficiency, high-accuracy fabrication of window, door, and facade systems where both UPVC and aluminium profiles are used, often in the same project or within a single manufacturing facility.

Main Uses and Applications:

1. **Automated Precision Cutting of Both UPVC and Aluminium:**
- This machine is engineered to handle the distinct material properties of both UPVC and aluminium, often with interchangeable or specialized blades and optimized cuttingparameters for each material.
- It provides precise mitre, square, and compound angle cuts essential for frame construction.

2. **High-Volume Production of Fenestration Components:**
- The combination of double heads (allowing simultaneous cutting of both ends of a profile) and CNC automation makes it ideal for large-scale manufacturing.
- It can produce hundreds or thousands of precisely cut profiles per shift with minimal manual intervention.

3. **Complex Geometric Cuts and Customization:**
- CNC control allows for the execution of intricate angles and profiles defined by computer-aided design software. This enables the fabrication of complex window and door designs, including those with non-standard angles or multi-point locking system preparations.

4. **Optimized Material Utilization:**
- CNC programming can optimize cutting sequences to minimize scrap material, leading to significant cost savings, especially when dealing with expensive aluminium extrusions.

5. **Consistent Quality and Reduced Errors:**
- Automated control eliminates human error associated with manual setting and cutting, ensuring every profile is cut to the exact specified length and angle. This leads to perfectly fitting frames during assembly.

6. **Reduced Labor Costs and Improved Safety:**
- Automation significantly reduces the need for skilled manual operators for the cutting task, lowering labor costs.

OPTIONAL ALUMINIUM –

- Coolant spray unit
- Hydro Pneumatic Cylinder
- Saw blade According to Aluminium
- Heavy mechanical Body Structure
- Motor Configuration

Component	Specification
Main Motor Power	1.5 kW × 2 (Total 3.0 kW)
Main Motor Type	Three-Phase Induction Motor
Servo Motors	1 units
Servo Motor Power	0.75 kW each
Pneumatic Pressure	0.6 - 0.8 MPa (6-8 Bar)
Air Consumption	70 L/min
Control Voltage	24V DC
Total Power Consumption	5 kVA

Performance Parameter	Specification
Cutting Angle Range	45 & 90 Degree
Standard Angle Accuracy	±0.1°
Cutting Length Accuracy	±0.1 mm
Repeatability	±0.05 mm
Cutting Cycle Time	10-20 seconds (both ends)
Maximum Cutting Speed	15-25 m/min
Daily Production Capacity	1000-1500 cuts (8-hour shift)

Parameter	uPVC/Aluminium Cutting
Blade Diameter	450 mm
Arbor Hole	30 mm
Blade Thickness	2.8-3.2 mm
Tooth Material	TCT (Tungsten Carbide)
Tooth Count	120 teeth
Tooth Geometry	ATB
Blade Speed	2800-3200 RPM)

9. DOUBLE HEAD AUTOMATIC 500MM SAW ALUMINIUM CUTTING MACHINE (ROXX-900A)

PRL
Precision.Reliability.Loyalty

A 500mm double-head automatic CNC aluminium cutting machine is a sophisticated piece of equipment designed for high-volume, high-precision cutting of aluminium profiles. Its primary applications and capabilities revolve around efficiency, accuracy, and versatility in manufacturing settings.

1. **Window and Door Manufacturing:**
- This is a core application. The machine is ideal for accurately cutting aluminium profiles used in the fabrication of window frames, door frames, sliding systems, and other architectural fenestration products.
- It can precisely cut mitre angles for corners and square cuts for other frame elements simultaneously.

2. **High-Volume Production Lines:**
- The double-head and automatic CNC features make it perfect for production lines where consistency and speed are paramount. It can significantly reduce lead times and increase output

3. **Simultaneous Double-End Cutting**
- Capability: The presence of two cutting heads allows for cutting both ends of a profile at a specific length simultaneously.
- Benefit: Dramatically reduces cutting time per component compared to single-head machines, increasing throughput.

4. **CNC Automation and Precision:**
- Capability: Computer Numerical Control (CNC) allows for precise programming of cutting lengths, angles, and sequences.
- Benefit: Ensures extremely high accuracy (often ±0.1 mm for length, ±0.1° for angles), leading to perfectly fitting assemblies, reduced waste, and consistent quality. Eliminates human error in measurement and setting

Parameter	Specification
Machine Model	ROXX-900A
Machine Type	Double-Head Automatic CNC Cutting
Primary Application	Aluminum Window & Door Profiles
Maximum Cutting Width	175 mm
Maximum Cutting Height	200 mm
Number of Cutting Heads	2 (Independent CNC Control)
Control System	Industrial PC-Based CNC
Machine Weight	1200 kg
Overall Dimensions (L×W×H)	2600 × 1600 × 1700 mm
Power Supply	380V/415V, 3 Phase, 50Hz/60Hz

Performance Parameter	Specification
Standard Angle Accuracy	±0.1°
Cutting Length Accuracy	±0.1 mm
Repeatability	±0.05 mm
Cutting Cycle Time)	8-15 seconds (both ends)

Component	Specification
Main Cutting Motor Power	1.5 kW × 2 (Total 3.0 kW)
Main Motor Type	Three-Phase Induction Motor
Servo Motor Power	0.75 kW each
Spindle Speed Range	0-2800 RPM
Pneumatic System Pressure	0.6 - 0.8 MPa (6-8 Bar)
Air Consumption	80 L/min
Total Power Consumption	6.0 kVA

Parameter	Specification
Blade Diameter	500 mm
Arbor Hole Diameter	30 mm / 40 mm
Blade Thickness	3.2-3.6 mm
Tooth Material	Tungsten Carbide Tipped (TCT)
Tooth Count	120 teeth
Tooth Geometry	TCG (Triple Chip Grind)
Maximum Blade Speed	2800 RPM

Component	Specification
Machine Frame	Heavy-Duty Steel Welded Construction
Guideways Type	Precision Linear Guideways HIWIN/PMI
Surface Treatment	Anti-corrosion Electrostatic Coating

System Component	Specification
Clamping Method	Pneumatic Automatic Clamping
Number of Clamps	8 Points (4 per side)
Clamping Force	800-1000 N per clamp
Clamp Stroke	80-100 mm
Profile Support	Adjustable Roller Support System

Safety Feature	Specification
Emergency Stop	4 Emergency Stop Buttons
Safety Guards	Polycarbonate with Safety Interlocks
Two-Hand Operation	Required for Cycle Start
Overload Protection	Motor Thermal and Current Protection
Air Pressure Monitor	Low Pressure Automatic Shutdown
Electrical Safety	Circuit Breakers, Fuses, Earth Leakage
Light Curtains	Optional Safety Perimeter

10. CNC UPVC & ALUMINUMS COPY ROUTER LOCK HOLE MACHINE (CRO-900)

The CNC UPVC and Aluminium Lock Hole Copy Router Machine is a highly sophisticated piece of equipment used in the fabrication of windows and doors made from both UPVC and aluminum profiles. Its main purpose is to accurately and efficiently cut precise openings (holes) for locks, handles, and other hardware components into these profiles.

Key Applications and Benefits:

1. **Precise Hardware Cut-outs:**

- Lock Mechanisms: Creates the exact shape and size of holes required for various types of lock cylinders, mortise locks, and their corresponding strike plates.
- Handles & Levers: Routinely produces openings for window handles, door levers, and espagnolette bolts.
- Hinges & Strikers: Can be programmed to cut out rebates and holes for hinges and strike plates.

2. **CNC Precision and Automation:**

- Computer Control: Operates based on digital designs (CAD files) loaded into the CNC system. This ensures extremely high accuracy and repeatability for every cut.

- Reduced Errors: Eliminates human error associated with manual measurement and routing, leading to higher quality finished products.
- Efficiency: Automates a complex and time-consuming process, significantly increasing production speed.

3. Versatility (UPVC & Aluminium):

- Material Adaptability: Equipped with appropriate tooling and spindle speeds, these machines can handle the different densities and hardnesses of both UPVC and aluminum profiles.
- Single Machine Solution: Allows manufacturers to use one machine for processing both types of materials, streamlining their operations and reducing capital investment.

4. "Copy Routing" Functionality:

- Template/Jig Based: Older or simpler versions might use a physical template or jig that the router bit follows. The CNC versions essentially digitize this process, allowing for much greater flexibility and complexity.
- Digital Programming: Modern CNC machines use software to define the path, shape, and depth of the cut, offering infinite design possibilities without needing new physical templates for each variation.

5. Optimized Workflow:

- Consistent Quality: Ensures that all cut-outs are identical, making the assembly of windows and doors faster and easier.
- Integrated Systems: Fits into a larger automated window/door fabrication line, working in conjunction with cutting, milling, and welding machines.

6. Surface Finish:

- Clean Cuts: The high-speed rotating router bit, guided by CNC, produces clean, smooth edges on the cut-outs, improving the aesthetic appeal and ensuring proper fit of hardware.

Parameter	Specification
Model	CRO-900
Applicable Materials	UPVC, Aluminium, PVC
Power Supply	380V / 415V, 50/60Hz,Three Phase
Input Power	2.6 kW – 3.5 kW
Motor Power	3 kW (High-speed models)
Rotating Speed	12,000 rpm (High-speed models)
Process Angles	0°, 90°, -90°
Copy-Routing Range	235×100 mm (Standard), 370×125 mm (High end models)
Cutter Diameter	8 mm, 10 mm
Working Air Pressure	0.5 – 0.8 MPa
Air Consumption	30 L/min
Machine Dimensions	870×1200×1500 mm (Compact),
Gross Weight	600 kg
Control System	CNC, Copy Routing, Automatic
Warranty	1- years
Key Features	Multi-hole drilling, water slot milling, handle hole milling, ventilation hole drilling

11. HEAVY DUTY COPY ROUTER LOCK HOLE & TRIPLE DRILL MACHINE -CRA600

PRL Make UPVC & Aluminium Lock Hole Copy Router Machine is a high-precision, high-speed 18000RPM Spindle motor routing solution designed for accurate lock-hole or keyholes, handle holes, and multipoint locking systems in UPVC window and door profiles.. Built for demanding fabrication environments, this machine delivers rapid, repeatable performance on both UPVC and aluminium profiles, ensuring flawless hardware alignment and enhanced assembly efficiency. The main use of a UPVC & Aluminium Lock Hole Copy Router Machine is to precisely and efficiently machine the necessary openings (lock holes, keyways, bolt holes, and other specific cutouts) into UPVC and Aluminium profiles for hardware installation.

- Lock hole patterning
- Accurately locate and mill standard lock cylinders, latch holes, and security hardware positions on UPVC/aluminium profiles and door/window hardware blocks.
- Reproducing existing hardware
- Copy a master template or sample to produce multiple identical lock hole patterns across parts.
- Through-bore and edge routing
- Create through-holes, rebate slots, and edge openings for

- for handles, multipoint locking systems, and escutche plates. High-repeatability cutouts
- Ensure consistent dimensions for multi-point locking systems, keeping alignment with hinges and webbing.
- Finishing prep for hardware installation
- Deburr, chamfer, and clean holes to facilitate smooth hardware fitment.
- Common capabilities (typical)
- 12- Multiple spindle options: fixed or plunge routers, some with variable speed
- 13- Adjustable guides/fences for different profiles and door/window thicknesses
- 14- Material compatibility: tailored for UPVC (PVC-U) and aluminium profiles
- 15- Finish quality: clean edge cuts with minimal chipping; optional pre/post countersinking
- 16- Optional dust extraction and chip management

Key Applications

- **Lock Hole Milling:** Creates precise rectangular or round lock holes in UPVC and aluminium profiles for installing window and door locks.
- **Hardware Fixing Holes:** Drills holes for handles, hinges, and other hardware, ensuring correct alignment and fit
- **Triple Hole Drilling:** Some models can drill three holes in a row, commonly used for multi-point locking systems .
- **Rebating and Milling:** Performs rebating (grooving) for hardware installation, typically up to 10mm depth .
- **Template-Based Copying:** Uses templates and copier pins to replicate hole patterns accurately across multiple profiles, ensuring consistency in mass production.

Technical Specifications:

Feature	Specification
Motor Power	3.5 kW + 1.5KW
Operating Pressure	0.6 – 0.8 MPa
Spindle Speed (Router)	18,000 RPM
Drilling Configuration	Triple (vertical) drill heads
Routing Template	Adjustable / Customizable
Power Supply	220V / 380V / 50Hz / 3 Phase

12. uPVC PORTABLE COPY ROUTER MACHINE CRO-300

The Portable UPVC Lock Hole Router Machine is specifically designed for creating precise and clean keyway and lock bolt holes in UPVC profiles for windows and doors. or custom-shaped recesses required to install locking mechanisms and handles onto uPVC window and door sashes and frames. Its portability offers significant advantages over stationary machines, allowing fabricators to work directly on assembled frames or in locations where moving large profiles is impractical.

Key Uses and Applications

- 1. Precise Lock Cylinder Holes:
- The primary function is to route the exact diameter and depth required for the lock cylinder. This ensures the cylinder fits snugly and operates smoothly.
- 2. Keyway Slots:
- It precisely cuts the elongated slot (keyway) through which the key is inserted into the lock mechanism.
- 3. Flexibility for On-Site or Workshop Use:
- Portability: Its lightweight and compact design allows fabricators to easily transport it around the workshop or even take it to installation sites.

- Working on Assembled Frames: This is a major advantage. Instead of disassembling a large window or door to take it back to a stationary router, this portable machine can be used to mill the lock holes directly onto the Portable End milling Machine
- 1. Fitting Handles: Routing the recess for the handle's base plate.
- 2. On-Site Fabrication & Repair: Essential for installing new windows/doors on a construction site or replacing hardware during renovation and repair work.

Component	Specification
Motor Power	1000 Watts
Collet Size	8 mm
Speed (No-Load)	25,000 - 30,000 RPM
Plunge Depth	Adjustable, typically up to 30 mm
Weight	60 kg

13. uPVC GLAZING BEAD CUTTING MACHINE - GLB300

A uPVC Glazing Bead Cutting Machine is a specialized piece of equipment designed to accurately cut the UPVC glazing beads to the precise lengths and angles required for fitting into window and door frames. Glazing beads are the strips that hold the glass or insulated glass unit (IGU) in place within the frame.
The UPVC glazing bead cutting machine is primarily used for precisely cutting UPVC or PVC glazing beads at 45-degree miter angles to ensure a perfect fit of glass panes within window and door frames. It can cut multiple bead pieces simultaneously, enhancing production efficiency

Precise Angle Cutting (Mitring):

- This is a critical function. The machine cuts the ends of the glazing beads at specific angles, most commonly 45 degrees, to create neat and tight corner joints when they are installed on the frame.
- Efficiency for Mass Production
- For manufacturers producing windows and doors in volume, these machines significantly speed up the process of preparing the glazing beads compared to manual cutting methods
- Equipped with multiple high-speed saw blades that provide clean, accurate cuts and often trim the trailing edge of the beads for

- seamless installation .
- Utilizes pneumatic clamping systems to securely hold glazing beads during cutting, minimizing vibration and improving accuracy
- Adjustable cutting speed to handle different bead materials and thicknesses .
- Comes with conveyor systems and adjustable supports to accommodate long bead profiles for smoother workflow.

Parameter	Specification / Details
Model	GLB-600
Input Power	1.5 kW
Voltage	380V, 50 Hz, 3-phase
Saw Blade Diameter	200 mm
Cutting Height	29 mm to 33 mm
Cutting Width	100 mm to 120 mm
Air Pressure	0.5 to 0.8 MPa
Air Consumption	Approx. 10 to 20 L/min
Spindle Speed	Around 2800 rpm
Machine Dimensions	Approx. 700×1000×1350 mm
Weight	300 kg
Cutting Angle	Automatic 45-degree mitre cuts
Additional Features	Pneumatic clamping, horizontal moving table

14. uPVC 3 AXIS WATER HOLE DRAIN SLOT MACHINE - DGA600

The UPVC 3 spindle motor water slot (drain slot) hole machine is specifically designed for milling precise drainage slots & holes in UPVC window and door profiles. These slots allow water to escape from window frames, preventing water accumulation ensuring long-term durability and performance of the window system
The UPVC 3-Axis Water Slot Machine is a high-precision tool designed to mill water drainage slots on UPVC window and door profiles. Built for speed, accuracy, and reliability, this machine ensures clean and consistent water slots across all frame and sash sections, enhancing both product quality and performance.

Drainage System Creation

- Water Drainage Slots: Cutting channels for water escape from frames
- Condensation Outlets: Creating paths for moisture evaporation
- Ventilation Channels: Air flow passages for pressure equalization
- Weep Hole Systems: Multiple drainage point preparation

2. Multi-Operation Efficiency

- Simultaneous Processing: Three operations in single cycle
- Different Slot Types: Various drainage patterns simultaneously
- Batch Consistency: Identical slot patterns across production
- Time Optimization: Reduced machining cycle times

Technical Specifications:

Feature	Specification
Axis Operation	3-Axis (Vertical + 2 Horizontal)
Motor Power	[e.g., 3 x 0.37 kW]
Operating Pressure	0.6 – 0.8 MPa
Slotting Speed	Adjustable
Profile Compatibility	Standard UPVC window & door profiles
Control System	Manual / Semi-Automatic

15. Heavy-duty End milling Mullion Machine MCA600

The UPVC Mullion Cutting & End Milling Machine is a specialized industrial tool designed for the precise fabrication of UPVC window and door components, specifically focusing on mullions. This machine combines two critical operations – cutting to length and end milling – into a single, efficient unit. This allows for the creation of accurate, ready-to-assemble mullion profiles, crucial for robust and aesthetically pleasing multi-lite window and door systems.

Key Features and Capabilities

This advanced machine offers a range of features to ensure high precision, efficiency, and ease of use:

1. **Dual Functionality:**
   - Precision Cutting: Capable of making clean, square cuts to exact lengths on UPVC mullion profiles.
   - End Milling: Precisely mills the ends of mullions to create necessary profiles for interlocking with transoms, sashes, or other frame components. This often includes creating specific notches or rebates.
2. **High Precision & Accuracy:**
   - Accurate Length Control: Features digital readouts or precise measuring stops for exact length cuts.

- Consistent Milling: Ensures uniform milling depths and profiles for perfect alignment and sealing in assembled units.
- Angle Cutting Capability (Optional): Some models may offer limited angle cutting capabilities for specialized designs.

3. **Safety Features:**

- Enclosed Cutting/Milling Area: Protects operators from moving parts and debris.
- Emergency Stop Buttons: Strategically placed for immediate shutdown.
- Safety Interlocks: Prevent operation when guards are not properly secured

Technical Specifications

Specification	Details
Milling Head Speed	High RPM for clean finishes - 2800 RPM)
Milling Motor Power	Varies (e.g., 1.5 kW)
Power Supply	Standard Industrial Voltage (e.g., 380V, 3 Phase, 50/60 Hz)
Pneumatic Supply	Required for clamping and some automation (e.g., 0.6 - 0.8 MPa)
Machine Dimensions	Varies significantly based on length capacity and features (e.g., 4m x 1.5m x 1.8m)
Machine Weight	500 kg
Clamping System	Pneumatic

16. MANUAL END MILLING MULLION CUTTING MACHINE UPVC & ALUMINIUM (MUC-300)

The uPVC and Aluminium End Milling Mullion Cutting Machine is a specialized industrial tool designed for precise milling and cutting of mullions—vertical or structural elements—in uPVC and aluminium profiles used in windows, doors, and frames. This machine ensures accurate end profiles, angles, and notches for seamless assembly, reducing errors and enhancing structural integrity.

Applications

1. **Milling of Mullions:**
   - Precisely mills the ends of uPVC and aluminium mullions, the vertical or horizontal bars between window or door panels, to prepare them for assembly.
2. **End Profiling:**
   - Creates accurate grooves, notches, or slots on mullion ends to facilitate fitting with other frame components like sashes, transoms, or corner connectors.
3. **Cutting and Trimming:**
   - Cuts mullions to specified lengths with clean, square edges for seamless frame construction.
4. **Hardware Preparation:**
   - Prepares mullions for the installation of locking

- mechanisms, hinges, or brackets by milling necessary holes or slots.

5. **Improving Fit & Finish:**

- Ensures tight, secure joints between mullions and other profiles, enhancing structural integrity and aesthetics of windows and doors.

6. **Versatile Material Handling:**

- Suitable for both uPVC and aluminium profiles, enabling flexible use in mixed material fabrication lines.

Parameter Category	Specific Parameter	Typical Range / Value
Processing Capacity	Material Compatibility	uPVC, Aluminium
	Max Profile Length Processed	1500 MM
	Max Profile Height	70 MM
	Max Profile Width	100 MM
Workpiece Handling	Clamping System	Pneumatic Clamps
	Pneumatic Pressure Required	0.5 - 0.7
Power & Utilities	Power Supply	380V/50Hz/3Phase (common)
Physical Specs	Machine Dimensions (L x W x H)	e.g., 3000 x 1800 x 1800
	Machine Weight	100
Safety Features	Emergency Stop Button	Yes
	Safety Guards / Enclosures	Yes

17. UPVC CNC 3 TOOL CORNER CLEANING MACHINE -CNC-300

The UPVC CNC 3-Tool Corner Cleaning Machine is an automated, high-precision piece of equipment designed to clean and finish the top & Bottom ,outer corners of UPVC window and door frames after they have been welded and cut. Its primary role is to remove excess weld material (flash) and create a smooth, clean internal corner radius, which is essential for both the aesthetic appeal and the functional performance of the finished UPVC product.

1. **Automated Precision Machining:**

- CNC Control: Being a CNC (Computer Numerical Control) machine, it allows for precise programming of the cutting paths, depths, and speeds. This ensures repeatability and high accuracy for every frame.
- Multiple Tools: The "3-Tool" designation indicates that the machine is equipped with three distinct cutting tools, each typically optimized for a specific part of the cleaning process:
- Tool 1 (Vertical Surface): Often used to clean the vertical weld bead that runs up the side of the corner.
- Tool 2 (Horizontal Surface): Used to clean the horizontal weld bead along the bottom or top of the corner.
- Tool 3 (Outer): This specialized tool carves the smooth, defined Outer radius at the junction of the vertical and horizontal surfaces.

- Workflow Process
- Loading: The welded uPVC frame is placed onto the machine's support table.
- Clamping: The pneumatic clamps are activated to secure the frame.
- Cycle Start: The operator selects the program and starts the cycle. The CNC then automatically:

Component	Specification
Control System	CNC with User-Friendly Touchscreen HMI
Number of Tools	5
Servo Power	0.5 kW of 2 Servo Motor
Motor Speed	2800 RPM
Positioning Accuracy	± 0.1 mm
Voltage	380V / 415V, 50 Hz, 3 Phase
Pneumatic Requirement	6 - 8 Bar (For Clamping)
Software	Pre-programmed for standard profiles, customizable for specials

18. uPVC CNC 5tool Corner Cleaning Machine CNC-600

The UPVC CNC 5-Tool Corner Cleaning Machine is an advanced automated system designed for the meticulous finishing of all 5 position in UPVC window and door frames. Building upon the capabilities of 3-tool machines, the 5-tool configuration offers even greater flexibility and precision in removing weld flash and creating sophisticated internal corner profiles. It removes excess weld material, burrs, and rough edges from all four corners, ensuring a smooth, professional finish on both the external and internal surfaces This machine is essential for achieving a flawless aesthetic and ensuring optimal sealing performance in high-end UPVC fabrication

Key Features

1. **Advanced 5-Tool System:**

- Versatile Cutting: Utilizes up to five different cutting tools, enabling specialized operations for flash removal, radius carving, and fine finishing on various surfaces.
- Profile Adaptability: Can be programmed to effectively clean and finish a broader spectrum of UPVC profiles and corner designs.

2. **Precision CNC Control:**

- High Accuracy: Ensures exact dimensions and consistent results for every corner cleaned.
- Programmable Operations: Allows for detailed programming of cutting paths, depths, speeds, and tool sequencing via an HMI.
- Repeatability: Guarantees identical finishing on every frame produced.

3. **Optimized Corner Radius:**

- Flawless Finish: Creates smooth, clean, and perfectly formed internal corner radii essential for aesthetics and gasket sealing.

4. **Efficient Weld Flash Removal:**

- Comprehensive Cleaning: Effectively removes excess weld material from vertical, horizontal, and junction areas.
- Automated Process: Significantly speeds up the finishing stage compared to manual methods.

5. **Safety Features:**

- Comprehensive Protection: Equipped with emergency stop buttons, safety interlocks on access doors, overload protection, and guards to ensure operator safety.

Parameter	Specification
Input Voltage	380V, 50Hz (customizable)
Input Power	2.5 kW
Milling Cutter Speed	2800 rpm
Air Pressure	0.4 – 0.8 MPa
Air Consumption	60 – 80 L/min
Profile Height	20 – 150 mm

Parameter	Specification
Profile Width	20 – 120 mm
Overall Dimensions	1800 × 800 × 2000 mm (typical)
Machine Weight	700 kg
Cleaning Time per Corner	15 – 25 seconds
Control System	CNC (Delta & Schneider)
Certification	CE, ISO

19. uPVC TOP & BOTTOM CLEANING GROOVING MACHINE –(GRV300)

The UPVC manual pneumatic top & bottom grooving cleaning machine is mainly used for cleaning and finishing the welded corners and grooved surfaces of UPVC window and door profiles. It removes excess weld material, burrs, and rough edges at the top and bottom grooves of the welded plastic frames, ensuring smooth, clean corners and enhancing the aesthetic and functional quality of the final product

Main Uses

- Cleaning welded corners: Removes rough weld beads and leftover plastic from the corners of UPVC window and door frames after welding.
- Groove cleaning at top and bottom: Precisely cleans the grooves on the profiles' top and bottom surfaces to ensure proper fit and smooth sliding of components.
- Enhancing product finish: Provides a smooth, uniform surface finish which improves the visual quality and mechanical fit.

- Preparation for assembly: Ensures the profiles are ready for hardware installation and assembly by removing imperfections caused during welding.
- Manual pneumatic operation: Pneumatic clamping and cleaning parts make operation safe, reliable, and easy without complex automation.

Parameter	Specification
Machine Type	Manual Pneumatic Groove Cleaning Machine
Operation Method	Pneumatic Power with Manual Feed Control
Cleaning Action	Top & Bottom Simultaneous Operation
Air Pressure Requirement	6 - 8 Bar (85 - 115 PSI)
Air Consumption	50 - 80 L/min
Air Connection	8 mm Quick-Release Coupling
Power Source	Compressed Air Only (No Electrical Power)

20. uPVC Interlock Punching Machine IPL-300

The UPVC interlock punching machine is used primarily for creating precise interlock grooves, slots, or holes on UPVC window and door profiles to enable smooth interlocking of sliding sashes or window/door components. This machine ensures accurate positioning and consistent punching necessary for fitting hardware that allows profiles to slide smoothly and lock securely

Key Uses:

1. Interlock Groove Punching: Produces slots/grooves on UPVC profiles where two window or door sashes overlap and lock together.
2. Hardware Preparation: Prepares holes or notches required for locking mechanisms or other fittings related to the interlocking system.
3. Smooth Profile Movement: Ensures the interlock system functions smoothly by ensuring precise machining of overlapping profile parts.[6]

Parameter	Specification
Machine Type	Pneumatic Interlock Punching Machine
Operation Mode	Manual
Punching Force	500N
Punch Diameter Range	12mm - 16 mm
Maximum Profile Width	50 mm - 120 mm
Maximum Profile Height	60 mm - 150 mm
Positioning Accuracy	± 0.1 mm
Punching Speed	1-3 seconds per hole
Cycle Time (Per Corner)	10 - 30 seconds
Pneumatic Pressure	6 - 8 Bar (for Pneumatic Models)
Air Consumption	40 - 60 L/min
Machine Weight	20 kg

21. ALUMINIUM CORNER PUNCHING MACHINE PUC-300

The Aluminium Window Manual Punching Machine is a foundational tool used in the fabrication of aluminium windows and doors. Unlike its more advanced CNC or automated counterparts, this machine relies on manual operation to create essential slots, holes, or notches in aluminium profiles. These punched features are critical for connecting different sections of the window or door frame, ensuring structural integrity and facilitating assembly.

Key Uses and Applications

1. **Creating Basic Joining Slots:**
- The primary use is to punch simple slots or notches into aluminium extrusions where two profiles will be mechanically joined (e.g., at corners, or for connecting transoms and mullions).
2. **Punching Holes for Fasteners:**
- It can be used to create holes for screws, rivets, or bolts that will secure components together.
3. **Preparing for Gaskets and Glazing:**
- In some simpler window designs, it may be used to create notches that help in seating gaskets or securing glazing beads.

4. **Assembling Frames and Sashes:**
- The punched features provide the necessary alignment and interlocking points for assembling the main frame, transoms, mullions, and opening sashes.

5. **Facilitating Manual Assembly:**
- It ensures that the profiles can be manually fitted together precisely without the need for complex adjustments or extensive cutting.

Parameter	Specification
Machine Type	Pneumatic Punching Machine
Operation Method	Manual Lever with Mechanical Advantage
Maximum Punching Force	500 - 2000 kg
Material Thickness Capacity	1.0 - 3.0 mm Aluminium
Profile Width Capacity	30 - 150 mm
Profile Height Capacity	20 - 100 mm
Hole Diameter Range	5 - 10 mm
Slot Length Capacity	Up to 100 mm
Positioning Accuracy	± 0.2 mm
Lever Mechanical Advantage	15:1 - 25:1 Ratio
Machine Weight	150 kg
Base Dimensions (L×W)	400 × 300 mm
Overall Height	500 - 800 mm

22. UPVC AUTOMATIC V-CUTTING MACHINE SLV-300

A "V-cutting machine" in the context of UPVC window making typically refers to a V-Notching Machine or a V-Grooving Machine. Its primary function is not cutting in the sense of severing the profile, but rather creating precise V-shaped notches or grooves on the UPVC profiles. These notches are crucial for specific aspects of window fabrication, particularly for:

The UPVC window making V cutting machine is used for creating precise V-shaped notches or grooves in UPVC profiles, typically at 45° or 90° angles. This V-cutting is essential for preparing the edges of UPVC window and door profiles so they can be perfectly aligned and welded together at the corners during fabrication

- **V-Notching:** Simultaneous cutting of equal V-shaped grooves on both sides of a profile using two blades, ensuring perfect angle cuts for corner joints.
- **Preparing Profiles for Welding:** Creates accurate V-grooves that facilitate seamless corner welding in UPVC window and door manufacturing processes.
- **Improving Joint Strength and Appearance:** The V-cut allows for better melting and fusion during welding, resulting in stronger and cleaner corner joints.

- **Applicable on Various Profiles:** Used on UPVC, aluminum-plastic, or composite profiles for window and door fabrication[2][3].
- **High Precision and Efficiency:** Enables fast, accurate cuts to optimize production efficiency and quality in industrial settings[2]

Specification	Detail
Blade Diameter	250 mm (2 blades)
Cutting Depth	20–60 mm
Spindle Speed	2800 RPM
Power	2 × 0.75 kW
Air Pressure	0.8 MPa
Air Consumption	80 L/min
Machine Dimensions	1300 × 700 × 1350 mm
Weight	230 kg

Parameter	Specification
Air Pressure Requirement	0.6-0.8 MPa
Air Consumption	50 L/min
Air Cylinders	3 Units (Clamping + Cutting)
Clamping Force	300-600 N
Filter System	3-in-1 FRL Unit

Parameter	Tolerance
Cutting Depth Accuracy	±0.1 mm
Cutting Angle Accuracy	±0.5°
Positioning Accuracy	±0.1 mm
Repeat Positioning	±0.05 mm
Surface Finish	Ra 3.2 µm

23. UPVC ARCH BENDING MACHINE (ARC-600)

The UPVC arc bending machine is used to form curved or arched shapes in UPVC/PVC profiles, primarily for manufacturing curved window and door frames. It can bend profiles into shapes like "C" and "U" by securely clamping the UPVC sections and applying controlled heat and pressure to achieve the desired arc or curve without damaging the material.

This machine is essential for producing architectural elements such as arched windows, doors, or frames, enabling design flexibility beyond straight profiles. It improves speed, precision, and repeatability in bending UPVC profiles compared to manual methods. The bending process helps in customizing and enhancing the aesthetic appeal of UPVC installations in construction and building industries.

Key Applications and Benefits:

1. **Aesthetic Customization:**

- Arched Windows & Doors: The most common use is to create the curved top sections of windows (e.g., half-round, segmental, Gothic arch) or doors.

- Specialty Shapes: It enables the creation of non-standard, decorative curved designs that would be impossible with straight cuts alone.

2. **Uniformity and Precision:**

- Consistent Curves: The machine uses controlled heating and a precise bending mechanism (often using molds or jigs) to ensure that the resulting arc is smooth, repeatable, and matches the required radius.
- Accurate Dimensions: It maintains the profile's integrity while achieving the desired curvature, ensuring that subsequent hardware installation remains possible.

3. **Material Integrity:**

- Controlled Heating: The machine heats the UPVC profile evenly to a specific temperature. This makes the material pliable enough to bend without cracking, deforming excessively, or degrading.
- Gradual Bending: The bending process is typically gradual, reducing stress on the material and preventing brittle fractures.

4. **Efficiency and Productivity:**

- Streamlined Process: Compared to manual heating and bending methods, an arc bending machine offers a much faster and more efficient way to produce curved UPVC components.
- Reduced Labor: Automates a complex shaping process, often requiring less skilled labor for the bending operation itself.

5. **Integration with Fabrication:**

- Standard Profiles: It allows fabricators to use standard UPVC extrusion profiles and shape them as needed, rather than requiring specialized, pre-curved extrusions which are often more expensive and less readily available.
- Post-Bending Operations: Once bent, these profiles can be cut to length, have drainage/water slots machined, and hardware holes punched, just like straight profiles.

Technical Parameter	Specification / Details
Model	ARC-600
Power Supply	220V, 1Φ, 50Hz (BAM-3MHA)
Total Input Power	5 kW to 7KW
Heater Power	5 kW
Heating Temperature Range	90-130°C
Heating Time	10-15 minutes
Compressed Air Pressure	7 Bar (100 psi)
Air Consumption	200 Liters per cycle
Bending Diameter Range	500-1800 mm / 100 mm minimum
Profile Width (Max)	120-150 mm
Profile Height (Max)	70 mm
Roller Diameter (Max)	180 mm
Worktable Dimensions	2870×1220×800 mm
Heating Tank Dimensions	3800×500×800 mm
Machine Weight	750 kg
Bending Capacity	C and U type profiles
Clamp Type	Special Mechanical clamps
Control System	Manual
Heating Method	Hot air

24. UPVC AUTO REINFORCEMENT SCREWING MACHINE (SQM-600)

The UPVC auto screwing machine is used to automatically fasten screws into UPVC window and door profiles, primarily for fixing steel reinforcements inside the profiles during the manufacturing process. It features an integrated screw feeding system and automatic screw driving with consistent depth control, allowing fast and accurate insertion of self-drilling or self-tapping screws without the need for pre-drilled holes. This significantly improves efficiency and precision compared to manual screwing, with screw insertion times around 2 seconds per screw .

Key Applications and Benefits:

1. **Structural Assembly:**

- Joining Frames and Sashes: UPVC profiles are typically joined using reinforcing steel liners inside them. The auto screwing machine drives screws through the UPVC profile and into these steel liners, securely connecting them. This is essential for creating rigid window frames, door frames, and sashes.
- Attaching Hardware: While some machines focus on frame assembly, others can be used to quickly attach hardware components like hinges, espagnolette bolts, and locking mechanisms directly to the UPVC profiles.

2. **Increased Production Speed:**

- Automation: Replaces manual screwing, significantly speeding up the assembly process. An automated machine can drive screws much faster and more consistently than a human operator with a handheld driver.
- Reduced Labor Costs: One operator can often manage multiple auto screwing machines or oversee the automated process, reducing the need for a large assembly workforce.

3. **Precision and Consistency:**

- Accurate Screw Placement: The machines are typically equipped with guides, jigs, or programmable heads to ensure screws are driven into the exact pre-determined locations. This is critical for structural integrity and for ensuring that hardware fits correctly.
- Controlled Depth and Torque: Most auto screwing machines allow for precise control over screw depth and tightening torque. This prevents over-driving (which can damage the UPVC or the liner) or under-driving (which can lead to loose joints). This consistency is vital for product quality.

4. **Ergonomics and Safety:**

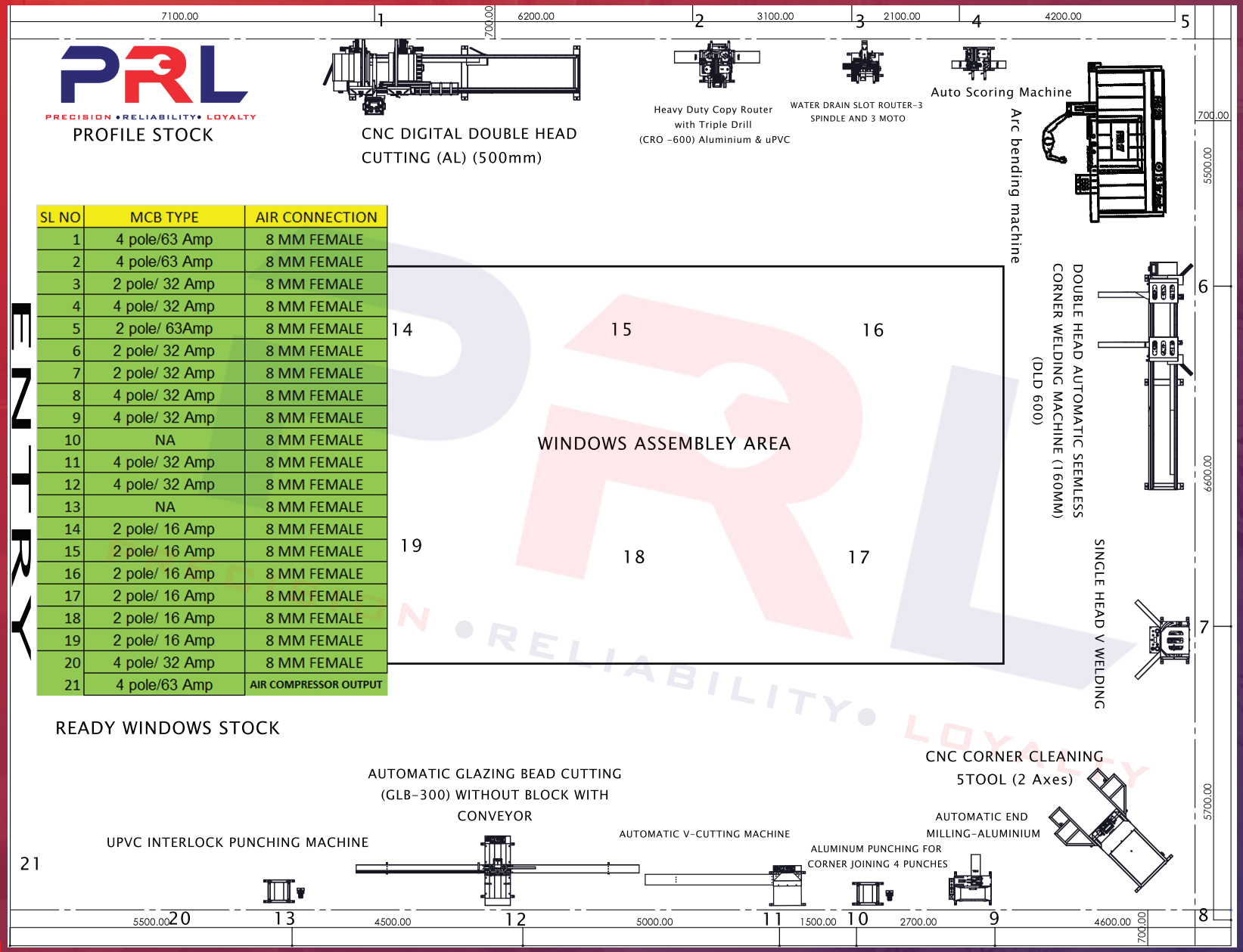
- Reduced Operator Strain: Eliminates the repetitive strain associated with manual screwing, improving worker comfort and reducing the risk of repetitive strain injuries.
- Controlled Operation: The automated nature of the machine ensures a safer working environment compared to manual tools where bits can slip.

- 5. **Versatility:**
- Adjustable Jigs: Many machines come with adjustable jigs to accommodate different UPVC profile sizes and types.
- Different Screw Types: Can often be configured to handle various screw lengths and diameters commonly used in UPVC fabrication.

Parameter	Specification
Screw Diameter Range	2.5 - 6.0 mm
Screw Length Range	10 - 80 mm
Screw Head Type	Pan, Flat, Round, Countersunk
Screw Drive Type	Optional
Screw Feeding	Automatic Vibratory Bowl
Screw Capacity	800-1000 pieces
Screw Recognition	Optical Sensor

Performance Metric	Specification
Screwing Speed	1.5 - 3.0 seconds/screw
Cycle Time	2 - 5 seconds
Production Rate	1200 - 1800 screws/hour
Torque Range	0.5 - 8.0 Nm
Torque Accuracy	±5%
Depth Control	±0.2 mm
Angular Accuracy	±2°

MACHINE ELECTRICAL



SL NO	MCB TYPE	AIR CONNECTION
1	4 pole/63 Amp	8 MM FEMALE
2	4 pole/63 Amp	8 MM FEMALE
3	2 pole/ 32 Amp	8 MM FEMALE
4	4 pole/ 32 Amp	8 MM FEMALE
5	2 pole/ 63Amp	8 MM FEMALE
6	2 pole/ 32 Amp	8 MM FEMALE
7	2 pole/ 32 Amp	8 MM FEMALE
8	4 pole/ 32 Amp	8 MM FEMALE
9	4 pole/ 32 Amp	8 MM FEMALE
10	NA	8 MM FEMALE
11	4 pole/ 32 Amp	8 MM FEMALE
12	4 pole/ 32 Amp	8 MM FEMALE
13	NA	8 MM FEMALE
14	2 pole/ 16 Amp	8 MM FEMALE
15	2 pole/ 16 Amp	8 MM FEMALE
16	2 pole/ 16 Amp	8 MM FEMALE
17	2 pole/ 16 Amp	8 MM FEMALE
18	2 pole/ 16 Amp	8 MM FEMALE
19	2 pole/ 16 Amp	8 MM FEMALE
20	4 pole/ 32 Amp	8 MM FEMALE
21	4 pole/63 Amp	AIR COMPRESSOR OUTPUT

MACHINE MENTAINANCE SHEDULE

Precision. Reliability. Loyalty

Regular maintenance is key to ensuring your uPVC machine operates efficiently & safely. Here's a breakdown of recommended tasks:

PREVENTIVE MAINTENANCE ACTIVITY

I	Profile Cutting Machine	Recommended Frequency
1	Cleaning-In/Out side of machine	Daily
2	Check FRL Unit oil Levels	Daily
3	Check the working presser (6to8 bar)	Daily
4	Cutting head alignment(45°/ 90°)	Daily
5	Check the Cutter Wheel edges	Weekly
6	To be check All Pneumatic valves, connecters and electrical points.	Monthly
7	Cutting Head Housing to be Clean Properly And Apply Oil	Monthly

II	Water Slot Machine	Recommended Frequency
1	Cleaning-In/Out side of machine	Daily
2	Check FRL Unit oil Levels	Daily
3	Check the Cutting Bit alignment	Daily
4	Check the Motor Angles alignments	Daily
5	Check the working presser (6to8 bar)	Daily
6	Greasing and oiling to the shaft if it's necessary	Weekly
7	To be check Pneumatic valves, connecters and electrical points.	Monthly

III	Copy Routing Machine	Recommended Frequency
1	Cleaning-In/Out side of machine	Daily
2	Check FRL Unit oil Levels	Daily
3	Check the Cutting Bit alignment	Daily
4	Check the Motor Angles alignments	Daily
5	Check the working presser (6to8 bar)	Daily
6	Greasing and oiling to the shaft if it's necessary	Weekly
7	Check the V-belt tension	Monthly
8	To be check Pneumatic valves, connecters and electrical points.	Monthly

IV	Welding Machine	Recommended Frequency
1	Cleaning-In/Out side of machine	Daily
2	Check FRL Unit oil Levels	Daily
3	Check the working presser (6to8 bar)	Daily
4	Clean the Heater Teflon cover	Daily
5	Welding, Cutting and Weld block alignment(45°)	Daily
6	Profile melting time, cooling time Stood be Maintain 30 Sec	Daily
7	Check the Heater Plate temperature (230°C to 250°C)	Daily
8	Change the Heater Teflon cover	Weekly
9	Greasing and oiling to the shaft if it's necessary	Weekly
10	To be check Pneumatic valves, connecters and electrical points.	Monthly

V	Corner Cleaning Machine	Recommended Frequency
1	Cleaning-In/Out side of machine	Daily
2	Check FRL Unit oil Levels	Daily
3	Check the working presser (6to8 bar)	Daily
4	Check the Cutters alignment	Daily
5	Check the Grooving cutter cutting edge	Daily
6	Check the Grooving cutter cutting edge	Daily

7	Check the Cutter cutting edges	Weekly
8	Greasing and oiling to the shaft if it's necessary	Weekly
9	To be check Pneumatic valves, connecters and electrical points.	Monthly

VI	End Milling Machine	Recommended Frequency
1	Cleaning-In/Out side of machine	Daily
2	Check the Cutters alignment	Daily
3	Check the Cutter cutting edges	Weekly
4	Greasing and oiling to the shaft if it's necessary	Weekly
5	To be check Pneumatic valves, connecters and electrical points.	Monthly

VII	Bead Cutting Machine	Recommended Frequency
1	Cleaning-In/Out side of machine	Daily
2	Check FRL Unit oil Levels	Daily
3	Check the working presser (6to8 bar)	Daily
4	Check the Cutter Wheel edges	Monthly
5	To be check All Pneumatic valves, connecters and electrical points.	Monthly
6	Cutting Head Housing to be Clean Properly And Apply Oil	Monthly

ADDITIONAL EQUIPMENT LIST

Essential Workshop Equipment's

These are must-have items for smooth production:

1- **Welding block die**
2- **Glazing Bead Block die**
3- **End Milling Cutter**

1.1 Work Benches & Assembly Tables

- Heavy-duty workbench (8–10 ft)
- Rubber-padded assembly table

1.2 Measuring Instruments

- Digital Vernier Caliper
- Measuring tape (5M / 10M)
- Angle meter / digital protractor
- Steel scale
- Spirit level

1.3 Hand Tools

- Screwdrivers set
- Allen keys
- Hammer + nylon hammer
- Utility knife
- Cutting plier
- Needle/nose plier
- Crimping tool
- File set

2. Additional Pneumatic & Power Tools

These save time and improve finishing quality:

2.1 Drilling & Screwing

- Electric drill
- Impact driver
- Pneumatic screw gun
- Magnetic drill (optional)

2.2 Hand Router (For Finishing)

- Electric trimming router with carbide bits
- End-milling bits
- Corner-cleaning bits

2.3 Air Compressor

- 5 HP – 500 LPM minimum
- Moisture separator
- Air hose pipe (quality PU)

3. Special Tools for UPVC Assembly

3.1 Glazing Tools

- Glazing bead cutter
- Bead puller tool
- Silicone gun (manual or pneumatic)
- Rubber mallet
- Glazing block set

4. Material Handling Equipment

- Profile storage rack
- Glass storage rack / trolley
- Finished window trolley
- Hand pallet truck (optional)
- Roller table for heavy profiles

5. Quality Inspection Tools

- Profile thickness gauge
- Welding strength tester (optional)
- Angle accuracy gauge
- Squareness checking jig
- Glass thickness gauge

6. Safety Equipment

- Safety gloves
- Eye protection
- Ear plugs
- Safety shoes
- Fire extinguisher
- First-aid kit

7. Office & Production Management Tools

- Computer with design software
- Job order printing system
- Production whiteboard

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